“It Seems Like Such a Simple Thing”: Challenges and Opportunities for Giving Credit in Science Museums

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Abstract—Science museums are a rich environment for visualization research where researchers have designed new visualizations, interaction modalities, and educational approaches. But they are also socio-technical places where power, context, and organizational structures influence how science communication is produced and perceived. In this paper, we draw on our collaboration with a local science museum, where we worked on creating a new interactive digital exhibit. Through this process, we encountered a socio-technical phenomenon: these exhibits rarely acknowledge the work or people behind them. We explored the reasons for this invisibility through an interview study with science museum staff and leadership. Our results reveal crediting of interactive exhibits to be a complex, situated challenge. This challenge, however, offers visualization research more than a practical problem to solve — it is an opportunity to reframe credit as a socio-technical design space, one where visualization can make visible not only the outcomes of science, but the processes and people that produce it. A free copy of this paper and all supplemental materials are available at https://osf.io/9vyu3.

Index Terms—science museums, feminist theory, interview study, input visualization, visualization design

1 Introduction

Science museums are informal learning spaces that engage the public with STEM topics [7, 21, 38], often with a focus on embodied learning through interactive and immersive experiences [57, 59]. The availability of scientific datasets and models, coupled with advanced visualization technologies, has provided science museums opportunities to bring new interactive experiences to visitors [72], as well as provided visualization researchers a context to directly engage the public. Visualization researchers have been involved with the design of a host of interactive exhibits, including the visualization of geological deep time [65], the microscopic biology of oceans [49, 50], and the tree of life [9, 39]. Other visualization research has focused on public-facing technical considerations, such as real-time, high-resolution rendering of large datasets [11] and the use of multi-touch tables [73]. And a slew of evaluative visualization research has studied how long people will engage with exhibits [40], how they decode visualizations [50], and how technologies like touch tables enhance learning [39].

As a team of visualization researchers, we had the opportunity to design a new exhibit for a local science museum in collaboration with museum staff. Towards the end of the design process, we thought about how to credit the people involved. Looking for existing practices within the museum, we found a striking discrepancy: while films shown in the museum routinely include credit reels listing contributors, interactive exhibits are typically presented without attribution. Approaching this discrepancy from a socio-technical perspective, and drawing on our academic backgrounds where authorship and credit are structured and expected, we understood this discrepancy not only as a practical issue, but also as a reflection of how some contributions remain invisible. This motivated us to examine how credit is understood, negotiated, and practiced in the context of interactive digital exhibits in science museums.

Our perspective builds on a stream of visualization research that has increasingly been studying visualizations not as isolated objects, but rather as artifacts situated within larger socio-technical contexts [1, 19, 24, 27, 54, 64]. Researchers emphasize that visualizations are not neutral representations, but are embedded within cultural, historical, and political conditions that shape how data is collected, structured, and communicated [6, 20]. This shift is reflected in work showing how people interpret visualizations through personal and social contexts [56], as well as in studies of real-world systems that highlight the role of organizational practices and infrastructures in shaping visualization production and use [2, 70]. Together, these perspectives direct our attention from what visualizations *show*, to how knowledge is produced: whose perspectives are represented, and which forms of labor remain unseen or undervalued.

Using feminist visualization perspectives as our analytic lens, we undertook an interview study to understand the absence of crediting practices for interactive exhibits in science museums. We interviewed a diverse range of people involved with various aspects of producing interactive exhibits, examining how they understand and experience existing crediting practices, how they encounter recognition or invisibility in their work, and how institutional structures organize contribution and authorship across teams. Our investigations began from the feminist assumptions of hidden labor [24] and perceived scientific objectivity [32], but instead revealed a complex, entangled, and empathetic set of socio-technical challenges and considerations for developing visual science museum exhibits.

With this work, we seek to draw attention to science museums as rich yet underexplored contexts for visualization research, particularly with respect to socio-technical questions of labor, visibility, and knowledge production. More specifically, we contribute an interview study that asserts a consistent absence of visible crediting practices for interactive digital exhibits, exposing a gap in how contributions are represented in public-facing visualizations. We found that this absence is due to a host of cultural, organizational, and practical challenges that shape how contributions are ultimately recognized and communicated. We argue that this gap is an opportunity for designing new visualization methods to support museums in both *producing* and *visualizing* credits. Building on this perspective, we conceptualize credit as a form of missing data and outline recommendations for both visualization design research and science museum practice. In doing so, we argue for treating credit as an integral part of how interactive exhibits represent the production of scientific knowledge.

2 Project Context and Background

We are a team of visualization researchers situated within a research environment that includes the Visualization Center C, a public science museum in Sweden. The Visualization Center C is jointly run between the city of Norrköping and Linköping University, and receives over 200,000 visitors a year. Exhibits and experiences at the Visualization Center C are developed in-house and make extensive use of visualization technologies to engage visitors in science communication. Most of

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the content in the Visualization Center C is developed in collaboration with researchers from Linköping University, providing unique opportunities for researchers to engage directly in scientific communication and outreach. For visitors, this setup provides opportunities to engage with cutting-edge technologies and scientific topics.

The Visualization Center C, like similar **science museums** worldwide, is an informal learning space that teaches visitors about a range of STEM topics through interactive and participatory experiences. Historically, these museums emerged as primarily collection-based and object-centered institutions, showcasing scientific collections and instruments [7, 38]. During the late twentieth century, many of these science museums shifted toward hands-on engagement and dialogic forms of communication, contributing to what became known as science centers [57, 59]. Today, the boundary between science museum and science center is increasingly blurred, and the terms are often used interchangeably [8, 58].

Science museums engage visitors in STEM topics through myriad formats, many of which include visual representations of data and models. **Interactive exhibits** are installations designed for active visitor engagement through physical, digital, or hybrid interaction. Rather than positioning visitors as passive observers, these exhibits invite exploration, experimentation, and embodied participation [34, 40]. Many science centers also operate dome theaters — sometimes called planetariums — that screen immersive audiovisual productions, typically consisting of a 180° domed screen, surround sound, and 3D capabilities. **Dome productions** are often experienced by visitors as a scheduled film with pre-rendered visuals and pre-recorded narration, although they can also include live facilitation by an educator or scientist. Other experiences at science museums include facilitated workshops, live demonstrations, and temporary exhibitions.

Science museums typically employ a range of staff, including leadership and management roles, designers, developers, technicians, educators, researchers, and operational staff. This staff is responsible for a broad range of work, from securing funding to designing exhibits to maintaining infrastructure. In smaller museums like the Visualization Center C, responsibilities are shared across staffers. Individual museums often design their visitor experiences in-house, emerging from collaborative processes involving conceptual development, research translation, interaction design, media production, fabrication, technical integration, facilitation, and ongoing maintenance [51]. Dome productions, however, require more specialized work and are typically produced and distributed by a smaller set of museums. The Visualization Center C, for example, produces films for dome theaters that are distributed to museums across Sweden.

The origins of the work we present in this paper stem from our experience designing an interactive exhibit for the Visualization Center C that engages visitors with the question: What is data? [4] Our conceptualization and design process for the exhibit included Linköping University researchers and students, Visualization Center C staffers, and local community members. As we were wrapping up the design of the exhibit and thinking about how to acknowledge the labor of the myriad people involved, it was unclear to us how to do this in an interactive exhibit. We looked in the Visualization Center C to find out how other exhibits handle crediting, and found that none of them do. This absence caught our attention because dome productions created in the museum *do* include credit reels, naming many of the same people involved with the production of exhibits.

As researchers who make use of feminist and critical theories, we wondered: is exhibition work a type of under-valued and invisible labor in science museums [24, 67]? Is the invisibility of human production in science communication a god-trick meant to signal detachment and objectivity [32, 45]? And besides, isn't crediting just for visitors who are unlikely to spend time engaging with it anyway [40]? *(Spoiler alert: the answer to all these questions is NO.)*

Initially, our inquiry was framed as a local design concern: how might we design and introduce crediting practices within our own science museum context. However, as our investigation expanded, it became clear that the absence of credit was not an isolated institutional issue but rather a widespread sector norm. Our attention shifted to questions of labor, power, responsibility, and inclusion around the design and deployment of interactive exhibits, both in our local environment and in science museums more broadly.

3 RELATED WORK

Our work sits at the intersection of two bodies of visualization research. We first review work that has positioned science museums as environments for visualization design, deployment, and evaluation. We then turn to a growing body of work that studies visualizations as sociotechnical artifacts — shaped by power, context, and the conditions under which they are produced. Our work addresses the intersection of these two subfields, presenting a largely underexplored topic: science museums are rich sites for studying how visualizations are situated within a public context.

3.1 Visualization Research in Science Museums

While visualization research has traditionally focused on supporting analytical insight and expert reasoning, today, a growing body of research explores visualizations in public spaces. Science museums are one such context, where visualizations mediate encounters between scientific data and museum visitors with the intention to engage, inspire, and educate. A string of work positions science museums as public-facing visualization environments, where visualizations are embedded within larger exhibition infrastructures [9, 49, 65]. In these settings, the visualizations are designed to accommodate the physical and social conditions of the museum environment, including walk-up interaction, short engagement times, and diverse audiences who explore content freely rather than following structured tasks [71]. These characteristics make science museums a unique context for visualization design and deployment.

A substantial body of visualization research focuses on the design of interactive systems for science museum environments, exploring how to support engagement, learning, and exploration with scientific content through interfaces adapted to the informal, social, and time-constrained conditions of science museums [9,39,49,50]. This includes the design of interaction techniques, narrative structures, and visual encodings that are accessible and meaningful to broad audiences. Prior work also examines how visualization design supports the translation of complex scientific content into public-facing narratives. This includes how storytelling and animation can help communicate unfamiliar scientific domains — such as metagenomics — to general audiences, though conveying complex causal relationships remains a persistent challenge [23]. This related body of work also includes investigations into the role of annotation and layout choices, such as time axes and labels, in shaping visitors' understanding of scientific diagrams, like evolutionary trees [52].

Alongside this work, visualization researchers have developed technical systems and infrastructure to support advanced visualization techniques within science museums. Platforms such as OpenSpace enable real-time exploration of large-scale astrophysical data in immersive dome theaters, supporting both guided storytelling and interactive navigation [10, 11]. Similarly, interactive volume rendering systems designed for touch tables bring a host of scientific datasets to the public [73]. These systems demonstrate how visualization can translate complex scientific datasets into engaging, embodied experiences for diverse audiences.

A further line of work focuses on the evaluation of visualization systems in museum contexts, examining how visitors engage with and interpret visualizations. These empirical studies show that both interaction techniques and visual content shape visitor engagement and understanding [39, 41]. Research on visualization literacy further highlights how museum exhibits support visitors in making sense of data, often through scaffolding interpretation and enabling participatory engagement with visualizations [60]. Interaction itself plays an important role in communication, where gestures, navigation, and presentation styles shape how meaning is conveyed to audiences [68].

Taken together, this body of work establishes museums as important environments for visualization research — spanning design, system development, and evaluation. Crucially, visualization plays a central

role in enabling public engagement with scientific knowledge. However, while visualization research has extensively explored how to design, build, and evaluate systems for museum audiences, it has largely left unexamined the socio-technical conditions under which these systems are produced. Museum visualizations are created by interdisciplinary teams — designers, scientists, programmers, educators, and technicians — and yet, questions of authorship, contribution, and crediting are ignored in visualization research, leaving the structures and practices that create visualizations largely invisible and understudied.

3.2 Visualizations as Socio-Technical Systems

With these questions in mind, we turn to a recent shift in visualization that has turned its attention to the study of visualizations as objects situated within larger socio-technical systems. Informed by feminist and critical epistemologies — approaches that foreground power, situatedness, and the politics of knowledge production, asking not only what knowledge claims are made but whose labor produces them and whose interests they serve — this body of work draws attention to the ways that power and privilege shape visualizations, challenging notions that data and visualizations are neutral and value-free [19, 24, 26, 27, 45]. Falling from these works is the increasing need to study the historical, personal, and situated contexts that are entangled with visualizations to understand how visualizations are made to be used and used to become useful [1, 64]. This trend in visualization parallels similar epistemic and methodological shifts within HCI that draw on feminist [33] and entanglement theories [30] to design and research technology in their relevant, complex, and nuanced contexts. Introducing feminist and critical theories shifts researchers' attentions to the role of power, history, and other social contexts that shape and mediate visualization and technology use, recentering relationships between people, data, and technology as important sites of knowledge.

Researchers employ critical theories to both design and study visualizations. Feminist theories have been used to critique visualization designs, calling to attention how certain design choices like clean lines and the use of geometric shapes reinforce objectivity and distance from the designer who constructed the visualization [26, 35, 45]. These concerns are echoed by others who highlight the political power of visualizations as tools that can be used for empowerment, or, conversely, harm and manipulation [19, 27].

With these critiques in mind, several researchers have explored how to design new visualization experiences. As an example, Morini et al. present alternative interaction mechanisms that address power imbalances between visualization authors and viewers [55]. In other examples, researchers have applied feminist critiques to their interface designs. For instance, Lin and colleagues directly address the power imbalance between data and expert knowledge by building a prototype that visualizes expert knowledge alongside official datasets [48]. Klein et al. also take feminist principles to heart in the design of an interactive book [46], focusing on making labor visible, humanizing data, and engaging the reader in critical readings of visualization choices used throughout the scrolly-telling experience. Adding to these select examples, visualization researchers have worked with feminist theory to develop alternative interfaces that support critical readings of data [43].

As fruitful as feminist theory has been for generating new design paradigms in visualization, it has also been equally generative in tuning researchers' attentions to the complex, messy, and highly situated contexts in which visualizations operate and exist. Akbaba et al. [2] took a feminist care ethics approach to understand the imbalance of power dynamics, agency, and access to maintenance in visualization research collaborations. Others have examined the role of power in participatory data physicalization activities, leading to recommendations on how to balance power imbalances when working with data in participatory contexts [17]. In close relation to power, visualization researchers have applied a feminist lens to unpack the ways that visualizations are political objects [6] that should be designed with communities in long-term, sustained efforts [42]. The socio-technical considerations of feminist theory have also been used to understand the impacts of bringing new visualization capabilities into organizations [70]. Within these studies, feminist theories shift our perspectives toward the complex relationships between individuals, institutions, data, and visualization technologies.

Science museums not only employ visualizations as important technologies for communicating science, but they also present opportunities to demonstrate how scientific knowledge comes to be to the public. Although prior feminist visualization research demonstrates the entangled, messy, and complex life of visualizations, this work has yet to reach science museums. The work we present in this paper, however, is in dialogue with existing feminist perspectives on museums more broadly, including prior work examining how museums function as complex systems of knowledge production whose workings remain largely invisible to the public [62], and more recent work framing the interplay of technology, data, and power in cultural spaces as a rich design space for reconfiguring what museums are and could be through an explicit articulation of values [16]. From these perspectives, we look at the labor behind producing exhibits through the lens of power, norms, and values. From our interview study, we find a rich and understudied ecosystem of visualization work that lacks systematic practices for crediting labor, ultimately hiding the people and processes behind the visual exhibits.

4 Methods

Our initial research objectives at the start of this study were to better understand the needs and opportunities for designing visualizations to attribute credit for the interactive exhibits within our local museum context. We conducted a round of interviews with staff at the Visualization Center C to collect their thoughts and experiences with credit-giving and -receiving practices within the museum. At this point of our process, we conceptualized credit as data that could be easily collected and serve as inputs into a visualization. After analyzing the interview transcripts, we conducted further work towards developing design requirements for visualizing credit. But this is where we got stuck. Credit, it turns out, was not easily defined. It evaded a clear definition, distinct boundaries, and singular use cases.

During this period Ryan — the third author on this paper who has 35 years of experience working in the planetarium field in the context of both science museums and natural history museums — joined the Visualization Center C for a sabbatical. We discussed our project with him, and he related similar absences of credit for interactive exhibits at science museums across the US. We decided to put our design project on hold and to instead shift towards developing a broader understanding of the reasons behind the absence of credit giving, the challenges of giving it, and the impacts of these practices. To enrich our findings from the initial interviews, we leaned on Ryan's professional network to recruit several senior managers from large science museums in the US — all of whom have years of experience working with interactive exhibits — to participate in a second round of interviews. We analyzed these additional interviews within the context of our original findings, and used these two perspectives — one from staff working directly on and with exhibits, and the other from management overseeing the everyday work and practices of museum staff — to develop the set of findings we present in Section 5.

In the following subsections, we discuss our interview and analysis methods and provide relevant demographics of our participants. Throughout this section, we include ↗ deep links to study materials for supporting traceability of our work [63].

4.1 Interviews

We conducted two rounds of interviews for our study: Round1 with local staff at the Visualization Center C in Sweden and Round2 with exhibit managers from large science museums in the US. Round1 interviews took place in-person, while Round2 interviews were all conducted over Zoom. All interviews took place in English, lasted approximately one hour, and were voice-recorded with permission from the participants. The participants did not receive financial compensation. Our study was exempt from ethics approval based on guidelines from the Swedish Ethical Review Authority.

For both rounds, we conducted semi-structured interviews using the pair-interview method [3]. Kate enacted the role of the *driver*:

leading the conversation, engaging directly with the participant, and controlling the progression of the questions. Miriah served as the *navigator*: keeping time, asking follow-up questions, and taking notes. Ryan joined for the Round2 interviews as a secondary interviewer with professional connections to the participants.

Round1 interviews took place during December 2024. We recruited participants based on our knowledge of diverse roles within the Visualization Center C staff and asked initial participants to suggest other colleagues during the interview. In total, ↗ we invited 10 staff members to participate, and conducted 8 interviews based on their availability. These ↗ Round1 interviews focused on the personal experiences and perceptions of the participants around the crediting practices within the Visualization Center C and towards their own contributions, including their thoughts on the roles of crediting and visibility within their work environment. We piloted Round1 interviews with two graduate students in our lab to refine the questions, timing, and interviewer coordination.

We conducted the Round2 interviews after our attention shifted from local crediting practices at the Visualization Center C to the broader norms and standards across the science museum sector. This round of interviews took place during June 2025 with exhibit managers ↗ we invited through Ryan's professional network. We selected these managers due to their seniority, substantial experience working with exhibits, and professional visibility within the science museum community. All three managers that we invited agreed to participate. The Round2 interviews followed ↗ an interview guide and focused on sector-wide attitudes and practices around crediting of exhibits, and elicited stories and perspectives from the participants based on their experiences overseeing the design and deployment of a range of interactive exhibits.

After each interview, the two primary interviewers — Kate and Miriah— held brief reflective discussions about the interview as well as the interview process. These discussions enabled us to note interesting findings and emergent themes as we went along. Kate transcribed each interview recording using Whisper Transcription [15], and reviewed the transcripts to correct transcription errors. In our reporting of quotes from these transcripts in Section 5, we *tidy-ed* them up — correcting grammar and cutting out filler words — to facilitate reading. We have included ↗ (anonymized) un-tidy-ed quotes as part of our supplemental materials.

4.2 Participants and Anonymization

We wanted to establish a trustworthy and respectful relationship with our participants to give them space to share their personal stories and perspectives. Our primary anonymization concern was that participants would be able to identify each other within our reporting: our Round1 participants work together in a small, highly collaborative environment; and our Round2 participants (and some of those from Round1) are tightly connected through the professional science museum community. As such, we prioritized anonymity first and foremost *between* participants throughout our study and reporting.

During the interviews, we informed participants: of our anonymization and ↗ consent processes; that they could request removal from the study at any point; and that we would not reveal who else we interviewed to them or others. In this paper, we not only anonymize quotes, but we have also assigned random IDs to the participants and report their relevant demographics in aggregate. We emailed each participant their tidy-ed quotes that we report in this paper and gave them the ↗ option to remove any. All participants responded to confirm their quotes. One participant requested a minor revision to their quote, which we incorporated; all others confirmed their quotes as they were written.

The participants from Round1 report years of experience relevant to their current position as ranging from 4 to 20 years. Their primary roles at the Visualization Center C at the time of the interviews were: educator, UX designer, research engineer, technical manager, producer, and researcher. The Round2 participants reported between 23 and 28 years of experience within the science museum context. Their management roles at the time of the interviews were Director of Editorial, Director of Exhibits, and Senior Vice President of Exhibitions.

4.3 Data Analysis

We worked with the interview transcripts following a diffractive analysis approach described by Akbaba et al. [2]. Rather than focusing on patterns and similarities within the transcripts, our diffractive analysis prioritized tensions, overlaps, and differences between the interviews and our interpretations of them. Throughout the analysis, we attended to concepts central to our theoretical framing, including recognition, invisible labor, and institutional hierarchy. Rather than treating these as predefined coding categories, we used them as lenses through which to read moments of friction, contradiction, and alignment in the interviews. We treated our differences in interpretation as productive sites of inquiry.

The analysis occurred in two rounds, following each round of interviews. The Round1 analysis was conducted by Kate and Miriah, and started by dividing the interviews into two sets. For each set, we did individual, close readings of the transcripts, followed by a discussion of emerging observations, preliminary themes, and tensions between formal crediting policies and the experiences of our participants. Kate and Miriah brought their notes and reflections to a group discussion with Derya, where we co-developed an initial ↗ thematic map.

The analysis for Round2 proceeded similarly, only in this round our discussions focused on how the Round2 interviews refracted, reinforced, or complicated the Round1 findings. We conducted an integrative analytic pass across both rounds, identifying specific passages from the transcripts that spoke to the ↗ themes. This process enabled us to identify themes that cut across our local environment, as well as more broadly across the sector. We received feedback from Derya and Ryan on our final set of findings.

5 Interview Findings

Our interview analysis revealed several recurring themes in how credit is understood and practiced in science museums. Participants described a prevailing absence of credit for interactive exhibits, while also articulating the values credit can hold, the challenges that make it difficult to implement, and the partial practices through which recognition ephemerally occurs. We present our findings around each of these themes in this section, and discuss their implications in Section 6.

5.1 The Invisibility-Norm

Across all of our interviews, participants reported that interactive digital exhibits are rarely credited with the names of museum staffers who contributed to their creation. Rather than framing it as a missing practice, participants repeatedly reflected on the absence of credit as if it were an established tradition, or what one participant explicitly called *"the norm"*:

> P5: *... I guess the norm, in relation to installations at the [science museum where I contribute] or other places, is that you don't do [crediting]. It is the organization that is mentioned, and the people involved become invisible...*

This *invisibility-norm* – the standard practice of framing interactive exhibits as anonymous institutional outputs rather than as authored works created by particular individuals – was itself invisible to many of our participants. During several of our interviews participants remarked that they hadn't deeply considered the crediting of their work or their colleagues'. One participant, P8, expressed surprise at the start of the interview that we had booked an entire hour to discuss the topic of credit, and another, P3 remarked:

> P3: *It seems like such a simple thing, right? Giving credit... I hadn't even realized this was missing.*

Importantly, participants rarely framed this absence of credit as a contested issue but rather as something that did not have precedence within science museums. P8, who leads a production team at a large science museum reflected on this:

> P8: *I don't think most people here are particularly concerned about it. I think it's just always been this way, so I*

don't think people are especially upset about it or expecting something different. But it's interesting, because many art museums do credit the people who worked on an exhibit – especially the curator. That makes a lot of sense, since a particular perspective or point of view often drives what is being shown... But at [this science museum], and in many science museums in [this country], there isn't really a curator. Exhibits are led by an exhibits team, with advisors who provide the scientific voice of the exhibition, but they are not credited as curators. So the way science exhibits are developed and credited is a bit different from art museums.

Because this absence of credit operates as an unquestioned norm, the process of crediting lacks standardized structure for interactive exhibits. Participants described science museums as having no formal policy for crediting exhibits, no standard exhibition production pipeline that includes credit, and no designated responsibility for determining how credit should be assigned. As one participant explained:

> P4: *It's not that much extra work. It's just not clear. No one has taken responsibility to figure out where we should add it or what names to include.*

Furthermore, participants noted that interactive exhibition production rarely follows a clearly defined pipeline, compared to the production of other experiences like dome productions, making the assigning of credit difficult not only in practice but conceptually, since the boundaries of individual contributions are often fluid and poorly defined:

> P3: *When there isn't a set exhibition production pipeline or methodology, it's harder to assign roles and credit accurately.*

And yet, contradictions to the invisibility-norm abound. For example, one participant described conventions for giving credit to certain external partners:

> P8: *There are certain people we always credit. That includes working with an Indigenous partner or group: we always credit them and acknowledge their contributions. If we work with a scientist from another organization and they provide quotes or specific information for a panel, we credit them as a distinct person.*

Another participant noted crediting conventions around certain media formats:

> P7: *The planetarium shows have always been credited, and the website products are always credited. If you use any in-house photo, that's always credited. And by contract, we credit all outside photos.*

These contradictions were sometimes linked to expectations and established norms outside of the culture of interactive exhibits:

> P6: *For a typical exhibit, we don't credit the exhibit developer. But videos are different. At the end of a video, you'll see the people who shot the film, wrote the text, or developed the animation. It's easier in that format, and it doesn't really interfere with the experience. It's something people are used to seeing.*

Taken together, these contradictions show that credit already performs important institutional work within science museums: signaling authorship, demonstrating ethical practice, and aligning with disciplinary and media-specific conventions. However, this work is unevenly distributed. Our interviews revealed that while external collaborators, data providers, and certain media formats are sometimes visibly credited within exhibits, the people who design, build, and maintain interactive exhibits inside science museums typically remain unnamed. This suggests a pattern of invisibility that operates at the level of everyday practice rather than deliberate intention.

Our interviews further revealed that the invisibility-norm persists not because credit is considered unimportant, but because its absence is taken for granted and rarely examined. Without established practices for crediting interactive exhibits, contributors have limited experiential grounding for imagining alternatives. As a result, the contradiction between extensive crediting practices of some science museum experiences – like dome productions – and the anonymity of interactive exhibit work can persist without becoming a sustained object of critique. We explore this tension in the following section, demonstrating that although credit is treated as optional or impractical within the invisibility-norm, it nonetheless carries significant personal, career, and institutional value for those involved in creating interactive exhibits.

5.2 Value of Credit

As researchers, we went into the interviews with the mindset that the target audience for exhibit credits would be the general public. But what we heard from some of our participants who are directly involved with exhibit creation is that *"...for the public I don't think it matters"* (P4). Instead, P4 reflected on the personal satisfaction that would come from sharing their work with friends and family:

> P4: *[It would be] fun if my family comes over and I show [them], here's my name in the credits.*

Similarly, we heard from other participants that being visibly credited would not likely alter material conditions. However, it still carried an emotional significance:

> P1: *I don't think [giving visible credit] would really change anything. Of course, it would be nice to feel a bit artificially proud if my name were somewhere, but I don't really believe it would bring me anything. It wouldn't add an actual apple to my plate, I might say. Maybe it would make me a little proud. And the only reason I'd want that is if [my kid] came here and said, 'This is my [parent]'s work.' That's the only thing I think it might add.*

The linking of professional labor to everyday social recognition also came up with P2:

> P2: *There was a journalist who wrote about [the exhibit I'd been working on], it got some attention, and that felt good.*

The personal value of credit was also closely connected to feelings of responsibility and care for the work itself. One participant suggested that the prospect of being publicly named would impact the level of attention they brought to their work:

> P11: *I think that when I know my name is on something, I'm more careful with some design decisions. Maybe it's not ethical, but it's how I feel.*

P11 also noted that the absence of recognition can accumulate over time and begin to affect motivation and future engagement within the workplace.

Beyond the personal value of crediting, participants also described credit as shaping professional visibility, future job opportunities, and, for some, the possibility of negotiating better pay. In this sense, credit functions not only as recognition but also as a form of career infrastructure that can support mobility over time. As one participant noted:

> P1: *I personally don't care much about credit, but I think it really matters for people earlier in their careers when it comes to jobs and pay.*

For example, visible credit could create the possibility for professional contact:

> P4: *I think it would be nice if someone saw the credits, liked the work, and could figure out who made it and get in touch.*

Participants also highlighted the negative case: when exhibits carry no public authorship, contributors must rely on informal trust from others to substantiate their past work. One participant described this problem directly:

> P6: *...if you were to go out looking for another job and say, 'Okay, I need to show some examples of my work,' people would have to take it on faith that I [worked] on these exhibits or projects, because it's not written on the piece itself.*

Without visible attribution, exhibit contributors may struggle to demonstrate experience, build professional portfolios, or translate their work into future employment opportunities.

Participants also described how credit matters at the scale of institutions. For example, several participants discussed credit in relation to funding and sponsorship. Naming funders with specific titles and levels of prominence was described as a non-negotiable part of exhibition production shaped by contracts and institutional agreements:

> P6: *And this is just to say that for purposes of funding, there is a whole other layer of credits for that. And there's a lot of rules around [which funders] we credit and how we credit them and so on and so forth.*

Other accounts linked credit to scientific legitimacy. One participant described a case in which the exhibit itself was not credited, but the data sources used within it were explicitly acknowledged:

> P7: *There's no credit for the in-house or outside talent who made [the exhibit]. But there is a credit panel for the sources of the data... We wanted visitors to notice, 'This is real data. It looks like animation, but it's backed up by real data.' So it probably serves two purposes: first, to give appropriate credit to the sources of the data, whether contractually required or not; and second, to help visitors understand that this is data – data visualization – not just something we made.*

Participants also connected credit to the educational mission of science centers:

> P3: *Giving visibility shows that there are people behind this work – it's not magic. Science doesn't just happen; it's made by people.*

They also noted that the importance of such visibility may vary depending on audience positionality:

> P10: *It depends on age groups as well, of course. I think many people want to know who created what they are seeing. It's like going to an art museum – you want to know who the sculptor was; otherwise it can feel a bit pointless. It can be a really nice thing to see, but you still start wondering who made this. For some people, it's just names on a plaque and they won't give it much attention. But I think the information could be important to some people, for example to notice who these people are, whether they are women or men.*

Across interviews, participants described credit as something that matters in concrete ways, even in contexts where it is rarely practiced. Rather than being discussed only as an ethical principle or a matter of fairness in attribution, participants described credit in terms of its practical implications: how contributors relate to their work, how that work circulates beyond the institution, and how museums position themselves toward funders and publics. These accounts also raise questions about for whom credit is intended, and whether its value lies solely in addressing museum visitors or operates across multiple audiences.

In short, credit is valuable. But as we discuss in the next section, assigning credit is also a challenge.

5.3 Challenges

As the previous sections showed, interactive exhibits are typically presented without individual attribution because science museums lack policies, procedures, or clear responsibility for crediting. Our assumption when we started the interviewing process was that the lack of standards for crediting results in an invisibility of labor. What we found, however, was quite the opposite: that the lack of standards makes it difficult to decide who *not* to include.

A major source of this difficulty is the scale and diversity of labor involved in interactive exhibitions, making the invisibility-norm preferable to drawing arbitrary boundaries:

> P8: *For example, at [the large science museum I work at], I would say that nearly every department and hundreds of people are involved in the making of any exhibition. Everybody from electricians to electronic engineers to writers to the people doing animal care to the people doing the design work... I think in many ways, it's just the tradition to not credit all those people because it's easy to leave somebody off accidentally and offend them. If you credit all those people, are you crediting your funding partners? If you hire a vendor or a contractor, are you crediting them? Where does it end?*

Given this breadth of labor, participants described how attempts to account for contributions can quickly become difficult to manage. Credit lists were imagined as expanding rapidly, with decisions about inclusion and ordering feeling subjective and hard to defend:

> P4: *I feel like it would be really simple to add a credits list. But someone has to collect the credits... it could probably go up to the hundreds. And I feel like it's subjective how you make this credit thing. In movies they have a rigorous system for showing names in order, all the actors and so on. But here it's more like, how do you sort people by relevance?... I'm not sure.*

However, participants also worried that credit given too broadly would lose its value:

> P7: *The question is always where to draw the line... The bigger [the credit list] gets, the less helpful it is to the people whose creativity was involved in making this product. So, you know, if I'm going to [credit] the plexiglass manufacturer and the lumber mill and everybody else, I think I've watered down what I was really trying to do. Which was to acknowledge the people whose brain power really went into this.*

Another challenge that emerged from our interviews concerns the situated nature of interactive digital exhibits: they are dynamic objects embedded in time and practice. For example, exhibits at many science museums are revised and adapted over time:

> P6: *An exhibit that we have that's maybe ten years old or something, we might create a new version of it, someone else might be coming in and making changes or changing the design or even changing sometimes fundamentally the interaction... We'll still call it the same exhibit. It's this thing where it's not like creating a work of art that has a fundamental integrity to what it is.*

Additionally, interactive exhibits require maintenance by tech staff, presentations to the public by educators, and only fully come into being through use by visitors. One participant reflected on how this kind of work often remains unrecognized:

> P2: *I think it's a bit sad, to be honest... The people who are out there [doing the on-going work] get very little validation for what they do, and they are pretty often forgotten.*

Taken together, these accounts show how the situatedness of interactive digital exhibits complicates credit-giving practices. When exhibits are continually revised and sustained through different temporal modes of labor, it becomes difficult to locate a stable *who* to which credit can be attached. Furthermore, credit must be bounded in order to remain legible and meaningful, yet any boundary also produces exclusion. These tensions help explain why, in practice, not crediting at all often becomes the (safer) default response.

5.4 Ephemeral Crediting

Despite the prevailing invisibility-norm, participants described a range of ephemeral practices through which their, and their colleagues' contributions are recognized. These practices acknowledge the personal and career value of credit. For example, several participants described acknowledgments during exhibit deployment celebrations:

> P8: *We'll have a celebration or we'll have a staff event to celebrate [a finished exhibit] and invite both the people on staff who worked on it and the partners.*

> P6: *Often there will be a big [list in an internal news message]. It goes out to everybody in the museum... celebrating that we finished the project. And here is everyone that I can conceivably think of who contributed to this project, and thanks to those people. Because it's really important to just say, 'We know we didn't do this just the 5 of us or the 7 of us or whatever. We are really grateful to all the people that actually made this thing happen.'*

These acknowledgments are also done as part of presentations and social media:

> P6: *I think when people present about [an exhibit] internally, at a staff meeting there's usually a slide that has 75 names on it.*

> P9: *When I [post on social media about] these exhibitions then I usually credit the people that actually did the work... to make them feel appreciated.*

We also heard about recognition that comes from visibility to important stakeholders, such as funders:

> P9: *Getting some sort of recognition is fun when you speak with the donors... [your contribution] has to be very important to give those recognitions.*

Presentations to stakeholders – normatively preformed by one or a small number of people – can also reinforce institutional hierarchies among diverse contributors and contribution types:

> P10: *The person listed at the top of the credits often has the opportunity to meet [stakeholders], and may therefore present the [museum experience] as their own. However, many others have also made significant contributions without ever having the chance to represent their work. Simply because they do not get the opportunity to meet these people. While there is nothing wrong about being placed at the top of the credits, it carries a responsibility to acknowledge the many individuals whose efforts made the [museum experience] possible and who equally deserve recognition.*

The influence of institutional hierarchy was also described by another participant who talked about the challenges of crediting of their team when their exhibits win awards:

> P7: *You would think about, 'Is there a priority? What's my hierarchy of crediting?'... Within my department we have an exhibition department of 40 people and I am sitting on the top of that department. I am not working nearly as hard on any given exhibit as the people whose hands are much more on it. So if they have to put credits in some sort of order, do they put me at the top or me at the bottom? Do they leave me off? Do they put me on another place?*

Our participants described ephemeral crediting practices as standard and frequent, and as important sources of personal and career value. These practices reveal a myriad of meaningful opportunities for acknowledging the labor behind exhibits even though they are momentary and invisible to the public. But they also reveal a persistent challenge of ordering contributors – whether intentional or not – that can reinforce institutional hierarchies. Taken together, these accounts point to the possibility for thoughtful design work in how to credit contributors in both fair and meaningful ways.

6 Discussion

The study we report in this paper started from a design question, namely, how to include credits for an interactive exhibit in our local science museum. As we considered our (lack of) options, we used standard feminist critiques of science to speculate about why interactive exhibits are largely anonymous. Was labor invisible, unacknowledged, and undervalued? Or perhaps, was it a symptom of distancing to reinforce science as an objective authority on knowledge?

What we found instead was more complex than these questions led us to anticipate. Across both rounds of interviews, participants described an invisibility-norm that operates not as a deliberate choice but as an unexamined given — something that had simply always been this way. At the same time, credit is far from meaningless to them: participants spoke about its value, and what it would mean to show their work to family, to build a professional portfolio, to feel that their contributions were legible beyond the walls of the institution. They also described the real difficulty of giving credit to exhibits that involve a broad range of labor when the risk of leaving people off is brought into tension with the desire to ensure credits are meaningful. And despite the absence of formal credit, recognition is already happening through a range of informal and ephemeral practices that acknowledge contributions, albeit without making them visible to the public. In short, *labor within science museums is known and valued, but difficult to acknowledge in fair and meaningful ways.*

Our findings also indicate that credit is not intentionally (or unintentionally) withheld in order to make science appear more authoritative and objective through what feminist scholars refer to as the *god-trick* [32]. In practice, the absence of acknowledgments emerges from a tangle of practical and normative challenges: no clear crediting pipelines within exhibit development processes; no sector-wide standards for what constitutes meaningful contributions to exhibits; and no designated responsibility among staff for making crediting happen. And yet this does not mean that the ideal of authoritative, faceless science is unrelated to the invisibility-norm — rather, we suggest, it may indicate that this ideal is so deeply embedded in the culture of science communication that it no longer requires conscious intention to endure [12]. Instead, *the erasure of human labor from public representations of science persists because nobody has thought to question it.*

In this section we reflect on the implications of our findings, but do so from our situated perspective as visualization researchers. We explore visualization opportunities across three dimensions: the production of credit as data, its representation for multiple audiences, and the organizational conditions needed to make crediting practices possible. These considerations make clear that the question of giving credit cannot be reduced to a purely technical problem as it is also a question of how contributions are constructed, represented, and negotiated [29]. As such, we speculate that researchers from other communities might find different but equally productive entry points from our findings.

6.1 Producing Credit Data

Credit, it turns out, is not the kind of thing that is simply waiting to be recorded. Our participants had no difficulty describing the labor behind exhibits. They could name who was involved, characterize what different people contributed, and articulate why those contributions mattered. The challenge was not articulation, it was delimitation. As P7 put it, *"the question is always where to draw the line."* Including everyone risks producing a credit list so long it loses meaning; drawing a boundary risks leaving someone out and, in doing so, implying that

their work did not count. This tension — between making credit inclusive while also keeping it meaningful — meant that for many participants, not crediting at all felt safer than crediting imperfectly. Rather than a problem of formalization, then, we interpret credit in science museums as a problem of negotiation: there is no single right answer for who should be credited or how, and that answer must be worked out anew within each exhibit context.

From a visualization perspective, we argue that this negotiation problem opens an interesting design space. Rather than simply designing visualizations to *represent* credit, we suggest that visualizations could be used in its *production*. The process behind credit giving in the CLEAR lab offers one compelling example of negotiation from a feminist perspective [47]. We wonder what kinds of visualization tools and processes we could design to bring contributors together to collaboratively produce meaningful credit datasets.

Existing input visualization approaches offer a partial solution. *Input visualization* [14] shifts the role of visualization from representing existing data to supporting the active construction of new data. Additionally, recent work on knowledge externalization [61] and embedded annotation [44] suggests that input visualization approaches can take digital, physical, or hybrid forms. But these frameworks tend to assume that the structure of what needs to be collected is known in advance: a predefined schema, a set of roles to be populated, a form to be filled in. Our findings show that this assumption breaks down for interactive exhibits. Contributions are distributed across many people and many moments in time, roles are not fixed, and what counts as a contribution may itself be subject to ongoing negotiation.

We speculate that a shift towards thinking of data as design material [28], and where credit is not treated as a fixed dataset to be collected but as something actively constructed and reshaped through the design process itself, could push input visualizations in new and interesting directions. From this perspective, the task for an input visualization is not to provide a better collection mechanism, but to support the ongoing negotiation through which credit comes into being: flexible enough to accommodate different kinds of contributions, different institutional contexts, and the open-ended, never-quite-finished nature of exhibit work.

6.2 Visualizing Credit Data

Our findings point to a second, concrete design challenge: how to make credit visible on or near exhibits in ways that are meaningful and engaging, without unintentionally reproducing institutional, hierarchical structures. During our interviews, participants raised concerns about ordering and visibility — who goes first, who gets left off, and what any ordering implies about relative value. This suggests that the design of credit visualization is not simply a matter of adding a plaque or a credit panel, but requires rethinking how contributions are ordered, compared, and presented to visitors.

This design challenge opens up a broader set of questions about how credit might be communicated, explored, and situated in the museum context. Museum visualizations are designed to accommodate walk-up interaction, short engagement times, and diverse audiences exploring freely [71], which means that any design for credit must compete for attention within conditions that are not built for sustained reading. At the same time, prior work in cultural museums suggests that even when the labor behind an exhibition is deliberately foregrounded, most visitors remain unaware of it [36]. This raises the question of not only how credit should be represented, but how it might be encountered — integrated into the experience of the exhibit itself, perhaps, rather than appended to its edges.

The challenge of representing credit in science museums reflects the broader challenge of representing credit for collaborative, scientific work. In scientific practice, contributions are typically organized as ordered lists, a convention that has proven difficult to move beyond even when there is clear motivation to do so [22, 47, 69]. Systems such as the CRediT taxonomy [5] aim to formalize contribution roles more transparently, but continue to rely on list-based representations that sort contributors into fixed categories. These approaches organize credit in ways that are comparable and rankable, rather than capturing the relational, overlapping, and negotiated character of collaborative work — precisely the qualities that our participants described as central to how interactive exhibits are actually made. Rather than reproducing list-based structures, however, visualization might support relational, temporal, or non-linear representations of contribution, where credit can be explored and connected rather than ranked and finalized.

Several existing projects suggest what such representations might look like in practice. GitHub's contribution visualizations foreground participation as ongoing and temporal, showing activity over time rather than presenting a final ranked list of contributors [31]. Work by Dörk and colleagues explores more relational and exploratory representations of collaboration, where connections between contributors and outcomes become the primary structure rather than their ordering [25]. And projects such as the labor chapter of *DataXDesign* demonstrate how credit can be woven into the interactive experience itself, making the work behind a visualization part of how visitors encounter and interpret it, rather than something appended as an afterthought [46]. Together, we argue these examples illustrate that visualizing credit is not only about displaying attribution, but about shaping how the work behind exhibits is understood and experienced, opening up possibilities for more inclusive, flexible, and context-sensitive representations.

6.3 Credit as a Socio-Technical Problem

The previous two sections outline how visualization could support the construction of contribution data, and how visualization might represent credit in ways that move beyond ordered lists. But our findings suggest that technical solutions alone will not be sufficient. Our participants did not describe the absence of credit as a tooling problem, but as an organizational and cultural one: no policies, no pipelines, and no shared sense of who was responsible for making crediting happen. Credit-giving, we argue, is not primarily a technical problem. It is also a socio-technical one [12, 24], and taking that framing seriously requires drawing on a richer vocabulary than the term alone provides.

Work on infrastructure in collaborative settings offers one such perspective. Infrastructure has been described not as a fixed technical foundation but as something relational, it takes shape through use, becoming infrastructure for someone in practice, and its absence is often only noticed when it breaks down [66]. This extends to how the ecology of visible and invisible work is maintained through ongoing, distributed effort that rarely surfaces as named contribution [67]. In museum contexts, facilitators have been shown to perform infrastructure design work that shapes the conditions for public engagement, yet this work rarely receives recognition as such [37]. Our findings mirror this dynamic: the work of crediting is itself a form of invisible infrastructuring, where the absence of formal systems means the labor of maintaining recognition falls to informal, ephemeral channels. Using infrastructure as a lens, we suggest that designing for credit is not only a visualization problem, but a problem of building the organizational scaffolding that makes crediting legible and sustainable.

This organizational dimension extends beyond museums alone. Makerspaces and libraries share with science museums an emphasis on participation, facilitation, and hands-on engagement, and the unseen design work of educators in these spaces has been shown to suffer from similar tendencies toward invisibility [53]. Making such labor legible requires not only documentation tools but cultural shifts in how expertise and contribution are valued. Our findings resonate with this: participants described similar dynamics in which the ongoing, maintenance-oriented work of exhibit educators and technicians accumulates value without accumulating recognition. The strategies developed in these parallel contexts may offer productive starting points for science museums seeking to develop more systematic crediting practices.

Deciding what counts as a contribution and where to draw boundaries are precisely the kinds of challenges that point to a gap in industry-wide guidelines for credit-giving. We suggest that the adoption of crediting practices at individual science museums may benefit from a community-wide discussion and establishment of new norms. This community work could connect to broader calls for science museums to become more socially relevant, reflective, and aware of visitors [59]. We argue that this outward-facing commitment could equally extend inward,

recognizing the myriad people whose labor shapes the experiences visitors encounter.

Our findings also show that credit is already being given within science museums in multiple ways beyond the exhibit itself — through internal recognition, professional visibility, and personal or informal channels. This suggests that credit does not need to be reduced to a single form of display, but instead approached as a broader design space that can accommodate different contexts, needs, and audiences [36]. These ephemeral, internal credit giving practices could be extended and institutionalized, and also made part of community-wide guidelines for acknowledging the labor behind interactive exhibits.

Beyond internal crediting, our findings point to an opportunity to make the production of science more visible to the public. Authorship has been described as the primary currency of the scientific realm, driving recognition, career advancement, and accountability [18]. Yet even within science, credit is contested: as collaborative work grows more complex, questions of who contributed what, and how that should be documented and recognized, require deliberate institutional effort to resolve [18, 69]. This contestation points to a deeper question that Macdonald raises in the context of science museums: where, and by whom, is science actually constructed — by scientists, by institutions, by technologies, or by the public itself [51]? Presenting science without crediting its makers obscures this question, reproducing what has been called a god trick: the suggestion that knowledge comes from nowhere and no one [24, 32]. As one of our participants (P3) put it plainly: *"Science doesn't just happen; it's made by people."* Designing for credit can make this visible, positioning science museums not only as communicators of scientific outcomes, but as spaces that reveal how science is produced [59].

What these directions share is a common epistemic commitment: visualizing and establishing credit is not only about attribution, but also about making the processes of science more visible and understandable. This positions visualization as a means of engaging wider audiences with how scientific work is produced, negotiated, and made meaningful, rather than only presenting its outcomes. Credit, in this light, becomes not a finishing touch but a representational choice with epistemic consequences.

7 Limitations

Like all empirical studies, ours too comes with several limitations that shape how our findings should be interpreted by readers. As a qualitative interview study with a limited number of participants, the findings we report are necessarily descriptive of specific contexts rather than explicit generalizations. Furthermore, our recruitment of interviewees relied on our local context and professional networks, and thus was limited to employees from only 4 museums across 2 countries, reflecting a Western institutional bias. While this allowed us to develop in-depth insights into specific practices of crediting, the perspectives we report are situated within particular organizational and cultural settings. Our management participants, however, did report extensive museum experience, including previous positions at other museums as well as extensive engagement with colleagues and museums across the world through conferences and work visits. Thus, while we concede that crediting practices may take different forms in other science museum contexts, we speculate that these contexts would most likely reflect the findings we report in this paper. Our study also focused specifically on interactive digital exhibits – we did not systematically examine crediting practices for non-digital or lower-tech exhibits, which may involve different norms and constraints, and which represent an important direction for future work. Validating our findings across a broader range of museum contexts and exhibit types remains an important next step.

8 Conclusion

In this paper, we examine the absence of crediting practices for interactive digital exhibits in science museums, motivated by our own experience of confronting an absence of credit in our local museum context. Through an interview study with staff and leadership across science museums in two countries, we found that this absence is a norm industry-wide. Contrary to our initial assumptions, the challenge is not that labor goes unrecognized — recognition is already happening, but through informal and ephemeral channels. The challenge is delimitation: credit is difficult to give fairly when contributions are broadly distributed, roles are fluid, and any boundary risks leaving someone out.

Central to our argument is a reframing that credit in science museums is not merely a missing practice but a form of missing data, and visualization is uniquely positioned to help construct, represent, and communicate it.This reframing shifts the question from a technical one to a socio-technical one, opening a concrete design space for future visualization research. We outline opportunities for how visualization can support the production and representation of credit in ways that move beyond ordered lists, while recognizing that no technical solution alone will be sufficient without shared institutional norms and collective agreement. The socio-technical tensions our findings surface extend beyond visualization. Researchers from other communities might find different but equally productive entry points here. We argue that credit is not a finishing touch to be appended to an exhibit, but a representational choice with real epistemic weight — one that shapes how we communicate not only the outcomes of science, but the processes and people behind it. Science doesn't just happen. It's made by people. Making that visible is an important role for visualization.

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